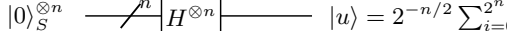
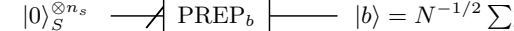
(a) $\mathrm{PREP}_S^{\mathrm{1BQF}} = H^{\otimes n}$



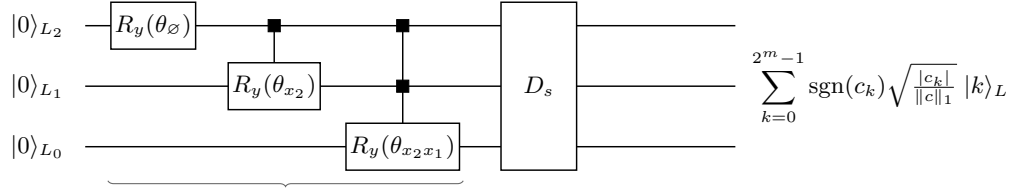
`one_bqf.py`: `h(b_qr)`. $\mathbf{b} = \delta\mathbf{1}$ padded with ones to 2^n, so the padding states are populated; they sit on $\lambda = s$ and the notch removes them.

(b) $\mathrm{PREP}_S^{\mathrm{QSVT}}$



`QSVT.build_circuit`: `StatePreparation(b_pad)`, $\mathbf{b}$ zero-padded: the $2^{n_s} - N$ padding states are never populated.

(c) PREP_L **and the sign diagonal** D_s **on the index register (**m **qubits;** $m = 3$ **drawn)**



A filled square is a uniform control: one angle per value of the qubits above. $\theta_v = 2\arctan\big(A(v_{\mathrm{R}})/A(v_{\mathrm{L}})\big)$ with $A(v)$ the norm of the amplitudes $\sqrt{|c_k|/\|c\|_1}$ under node v . $D_s|k\rangle = \mathrm{sgn}(c_k)|k\rangle$. The code uses `StatePreparation(amps)` and `DiagonalGate(signs)`; the cascade is what that synthesises to. Coefficients with $k > d$ are zero.